

ATLAS Bench and Ground Test Plan

General Test Ideas

General idea: Before ATLAS flies, each major function on the board should be tested on the bench or on the ground. The goal is to prove the sensors, logging, telemetry, pyro outputs, vibration durability, and basic flight logic before putting the flight computer in a rocket.

Board Function	General Test Idea
Barometer	Pressure pot test. Change the pressure around the board and confirm the barometer reads pressure and altitude changes correctly.
GPS	Outside test and hill test. Confirm GPS fix, latitude, longitude, altitude, satellite count, and altitude change while moving uphill or downhill.
LoRa	Distance test. Send telemetry from ATLAS to the ground station at increasing distances and check signal strength, packet loss, and reliability.
microSD	Fake log test. Run fake flight data and confirm ATLAS creates a log file, writes data, and saves the file correctly.
Flash	Backup log test. Store fake events or backup flight data, reset the board, then confirm the data reads back correctly.
Pyro	Multimeter and e-match test. First test with a multimeter and dummy loads. Then test with e-matches in a safe setup. Confirm main and drogue outputs only fire when commanded.
Vibration	Paint shaker test. Vibrate the board while logging data and confirm there are no resets, sensor dropouts, SD card issues, or false flight events.
IMUs	6-face measured orientation test plus live 3D viewer test. Place the board in known orientations and confirm the 3D model matches the real board movement.
Accelerometer	Tap test in each direction. Tap the board in positive and negative X, Y, and Z directions and confirm the correct acceleration axis spikes.

Possible Real Flight Validation Test

If everything goes well and the board is ready sooner than expected, ATLAS might be tested on the University of Ottawa rocketry team's G-class rocket.

We already have possible access to a professor's farm for flight testing. Before doing this, we need to confirm permission from the UO rocketry team, confirm the size of the farm, and make sure the selected G-class motor is safe for the available launch area.

The goal of this test would be to collect real flight data and validate the flight computer in an actual rocket flight before moving to larger flights.